

Bump'em carrier board: Teensy 4.1 + MCP4728 DAC -> 4 cable modules (A left, B front, C right, D back)

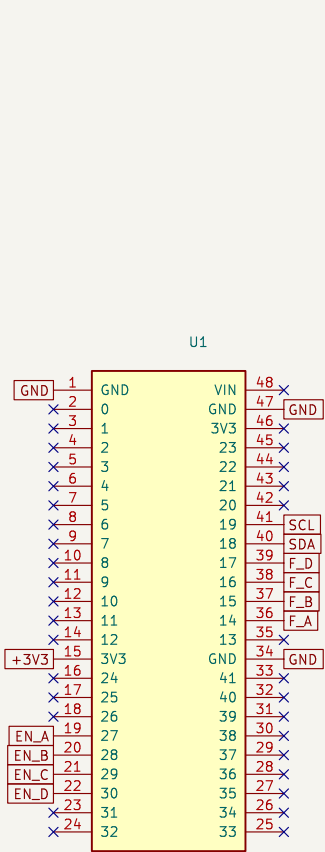
Pins as in firmware/src/main.ino and docs/diagrams/cabling-4-modules.png: enable T27-30, setpoint DAC VA-VD, force T14-17 (A0-A3). I2C T18 SDA / T19 SCL

Teensy VIN (5 V) is never connected. Teensy and DAC plug into female headers. DAC VCC fed on pad 12; pad 1 (VCC) is the same net on the breakout, left open here.

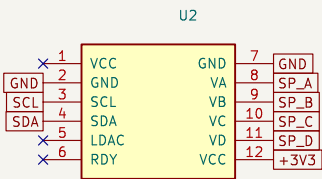
Force input protection per module: R1-R4 series (1k) | R5-R8 pull-down (1M, unplugged input reads 0 N) | D1-D4 BAT85 clamp to 3.3 V.

If the IAA100 output can exceed 3.3 V: R series + R pull-down form a divider (e.g. 0-10 V: 22k / 10k -> 3.1 V). Re-calibrate after any change.

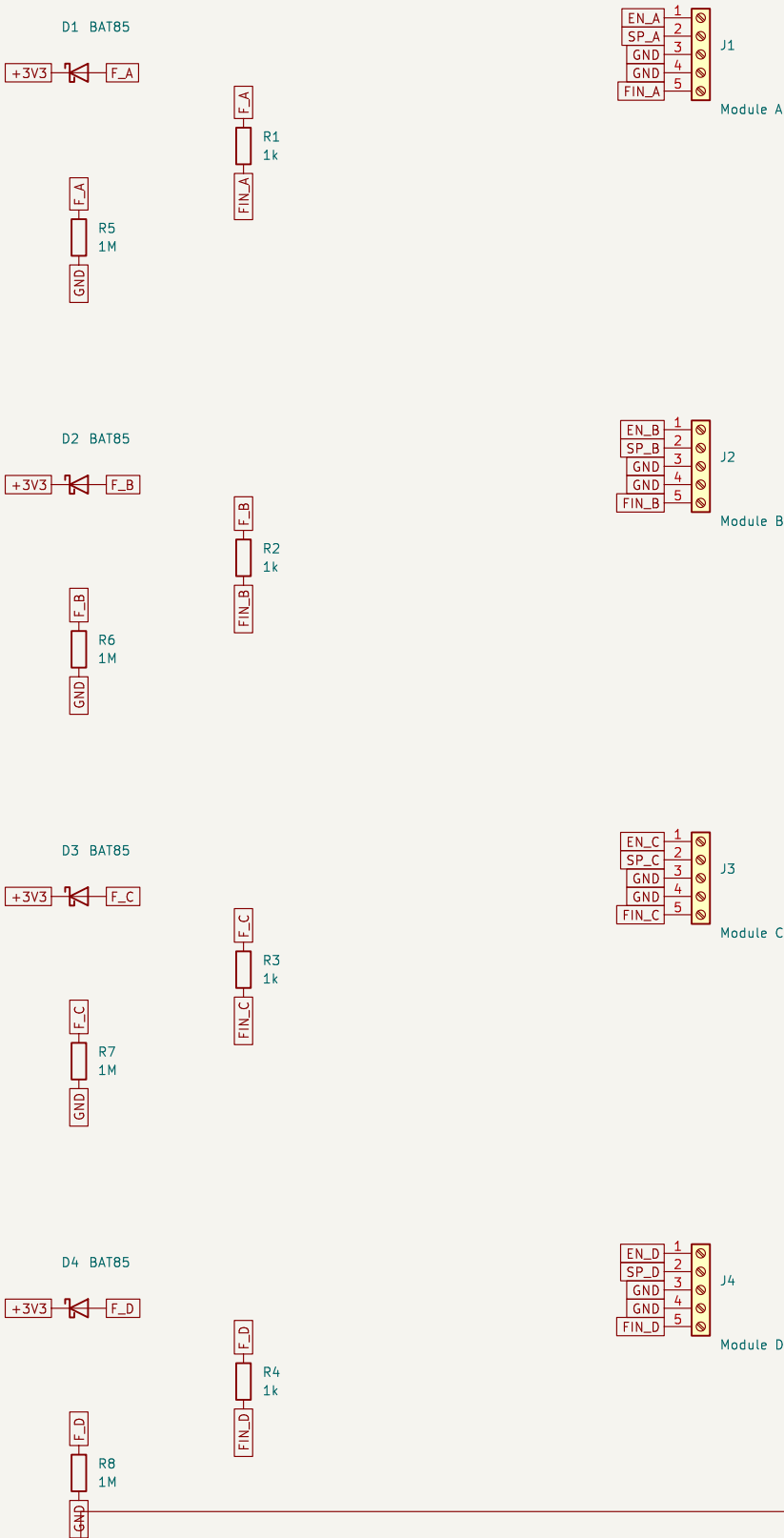
Terminal pins per module (wire colours): 1 EN yellow, 2 SP+ blue, 3 SP- green, 4 GND white, 5 F brown



Teensy 4.1



MCP4728 (Adafruit 4470)



Generated by hardware/carrier/gen_schematic.py from netlist.py; edit those, not this file
IBMS Offenburg

Sheet: /
File: bumpem-carrier.kicad_sch

Title: Bump'em carrier board

Size: A3 Date: 2026-10-02

KiCad E.D.A. 8.0.9

Rev: 1

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